



Exploring Global Competitiveness Index 4.0 through the lens of country risk

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ABSTRACT

The Global Competitiveness Index (GCI) 4.0 provides an annual evaluation of country-level productivity and development in the fourth industrial revolution era. The GCI 4.0 is classified into 12 pillars representing socio-economic, financial, health, technological, and other factors of production. Country risk, which is an aggregate measure of country-level exposure to financial, political, and economic risks, can influence national development. This study explores the association between country risk and competitiveness by mapping country risk drivers to GCI and GCI pillars to country risk. Two Bayesian Belief Network models, with a minimum prediction accuracy of 81.3 %, are developed using two data-sets representing the GCI and country risk ratings of 107 countries. Sensitivity analysis reveals that low performance in information and communication technology (ICT) adoption (monetary policy) can significantly affect country risk (GCI), whereas innovation capability (financial system risk) and institutions (legal system) can considerably improve country risk (GCI). By delving into how the pillars of competitiveness interact with various risk factors such as political stability, economic volatility, and regulatory environment, decision-makers, investors, and policymakers can gain valuable insights that guide effective resource allocation, risk mitigation strategies, informed policy formulation, and sustainable development efforts.

1. Introduction

Regional competitiveness represents the efficacy of structural economic policies in attaining sustainable development and productivity (Grassia et al., 2022; Im and Choi, 2018). The Global Competitiveness Index (GCI) 4.0 measures the national competitiveness of 141 countries in the fourth industrial revolution era (WEF, 2019a). The GCI 4.0 was introduced in 2018 in the Global Competitiveness Report, which has been published every year since 1979 (Sarabia et al., 2021). The GCI 4.0 guides policymakers and other stakeholders on policy choices related to long-term growth and holistic economic strategies (WEF, 2019a).

Using a bottom-up approach, GCI 4.0 is computed by aggregating scores from the indicator level to the overall score, including assessments of 12 pillars and 103 indicators (Sergi et al., 2021). The 12 pillars include institutions, infrastructure, information and communication technology (ICT) adoption, macroeconomic stability, health, skills, product market, labor market, financial system, market size, business dynamism, and innovation capability (WEF, 2019a). Almost half of the indicators are based on an annual Executive Opinion Survey conducted by the World Economic Forum (WEF), involving approximately 15,000 business executives across the world (Önsel Ekici et al., 2019). The country-level score of each question of the survey is the average score of

all answers for each country, implying that all answers carry the same weight for a given question.

Global businesses and financial markets have become more interdependent and vulnerable to global changes, necessitating a better understanding of country risk, which is contingent on the political, financial, and social stability of a country. Country risk can adversely impact the expected return on investment, business climate, and the performance of foreign firms (Shostya and Banai, 2017). The uncertainty of investors and firm managers increases with the increase in country risk, which increases the risk premium that investors require and decreases their tolerance for long-term investments (Husted and Sousa-Filho, 2017).

Business managers consider country risk as an important element of their strategic decision-making process because country risk can significantly impact the performance of an international corporate entity (Lee and Wang, 2021). However, country risk assessment is a challenging problem due to the complexity associated with modeling multiple interdependent macroeconomic, social, and political factors (Doupmpas and Zopounidis, 2001). In particular, corporate performance is significantly affected by financial risks that occur due to the changes in the economic environment, such as interest rates, exchange rates, and inflation (McDowell, 2021). Similarly, economic risk is contingent on a

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given country's macroeconomic policies. For example, a sound monetary policy and low inflation and unemployment rates contribute to reducing country risk. When a country's economic conditions become unstable, the country risk may increase (Oetzel et al., 2001). Political risk analysis involves the assessment of political events that may affect the operations and profitability of international businesses (Gregory, 2020). Political risk factors can be classified into macro- and micro-risks. Macro-risk results from politically motivated environmental changes that can impact all foreign enterprises, whereas micro-risk may only affect selected industries or firms in a country (Oetzel et al., 2001).

Several studies have explored causality among the GCI pillars (Cinicoglu et al., 2017; Kabak et al., 2020; Önsel Ekici et al., 2016, 2019) and country risk drivers (Qazi and Khan, 2021; Qazi et al., 2022) regarding GCI and country risk assessments, respectively. Exploring such cause-effect relationships among predictor variables can help identify the relative importance of the drivers of a target variable for resource allocation decisions (Delen et al., 2020; Simsekler and Qazi, 2022). To the best of the author's knowledge, the association between country risk drivers and global competitiveness pillars remains under-explored. Since country risk can significantly impact the growth and development of an economy (Cavusgil et al., 2020; Liu et al., 2020; Tang and Buckley, 2020), it is important to identify critical GCI pillars that can help national policymakers reduce country risk. Similarly, critical country risk drivers influencing competitiveness can be identified for resource allocation and improving national productivity.

The motivation for exploring the association between GCI 4.0 and country risk lies in the recognition that a nation's competitiveness is intrinsically linked to its exposure to various risks (Qazi and Khan, 2021). By uncovering the correlations between these two critical aspects, policymakers, businesses, and investors can uncover nuanced insights that inform strategic decisions (Qazi et al., 2023). This knowledge can enable governments to formulate targeted policies that enhance competitiveness while addressing vulnerabilities, empower investors to make well-informed choices by assessing risk-return profiles accurately, and aid in proactively managing potential challenges to foster economic resilience and sustainable development on a national level.

The main objective of this study is to identify critical GCI 4.0 pillars and country risk drivers influencing country risk and global competitiveness, respectively. This study explores a Bayesian Belief Network (BBN) approach, which can effectively model risk and uncertainty (de Oliveira et al., 2012; Ekici and Önsel Ekici, 2021; Lawrence et al., 2020). Utilizing data by AM Best (AMBEST, 2019) and WEF (WEF, 2019a), two probabilistic network models are developed to capture dependencies between country risk (drivers) and GCI 4.0 (pillars). The remainder of the paper is organized as follows: Section 2 presents a brief overview of the relevant literature. Section 3 delineates the research methodology and Section 4 describes the modeling results. Section 5 presents the implications of the results and Section 6 provides the directions for future research.

2. Literature review

2.1. Global competitiveness

The GCI encourages leaders to adopt a systems thinking-based approach to formulate and manage today's complex global problems (Kabak et al., 2020). Policymakers can develop holistic solutions and strategies by contextualizing different drivers of competitiveness and productivity as interdependent factors that belong to a vast system (Önsel Ekici et al., 2019). Several studies have emphasized the importance of competitiveness in sustaining economic growth and development. For instance, Pal et al. (2022) found that the GCI can explain the economic well-being of a nation through attracting more investment. Similarly, Athari et al. (2020) reported a positive impact of competitiveness on capital flows, with the strength of association contingent on

the specific stage of economic development and the country risk level. However, the GCI pillars and associated drivers must be prioritized based on their impact across different sectors. For instance, Sergi et al. (2021) established the importance of strategic GCI sub-components, namely infrastructure, human factor, and institutions, in improving logistics performance. While all the identified variables play a crucial role in facilitating comprehensive sector development, specific variables or clusters are dominant with respect to regional economic advancement, a pattern that tends to align with geographical distribution across continents. Similarly, Dima et al. (2018) identified innovation and education as critical determinants of competitiveness and economic development of the European Union.

Limited studies have explored the association between GCI pillars and other variables in a network setting. Exploring network-based modeling can help establish synergies and trade-offs across a network of interacting variables (Dawes, 2022; Xiao et al., 2023). For instance, Önsel Ekici et al. (2019) analyzed the association between the GCI pillars and the Logistics Performance Index (LPI). They identified some critical pillars, including technological readiness, higher education and training, innovation, market size, and infrastructure, that can help improve the country-level logistics performance. Similarly, Kabak et al. (2020) investigated the two-way interaction between the competitiveness and logistics performance of countries using a BBN model and identified the most critical GCI pillars as business sophistication, financial market development, infrastructure, goods market efficiency, and higher education and training. Using an Artificial Neural Network (ANN) model and a Cumulative Belief Degrees (CBD) approach, Önsel Ekici et al. (2016) established a close relationship between country-level global competitiveness and logistics efficiency. The utilization of network-based models in these studies can assist policymakers in strategically determining priority competitiveness pillars to allocate resources effectively.

The Global Information Technology Report released by the WEF has introduced the Networked Readiness Index (NRI) to measure the global competitiveness of a country's ICT diffusion. Using the partial least squares path modeling approach, Wu et al. (2012) explored the critical pillars within the NRI and established the causal relations among them. Business usage, business readiness, and market environment were identified as the three critical pillars to manipulating the NRI overall scores, whereas government readiness was the foremost enabler. Based on the results, policymakers may allocate limited resources to the three critical drivers and one foremost enabler to achieve competitiveness of national ICT diffusion.

2.2. Country risk

The nature of the interdependent and dynamic constituent factors makes the country risk assessment more challenging (Li et al., 2012). Brown et al. (2015) introduced a new metric for country risk assessment with a focus on governance, economics, operations, and society. Several credit rating agencies, such as Moody's, Standard and Poor's, and Fitch, have been regularly publishing their assessments of the risk associated with an investment in various countries. Country risk ratings have a direct impact on the cost of borrowing as they represent the country-level probability of debt default. An increase in country risk will increase a country's cost of borrowing and debt servicing obligations (McAleer et al., 2011). Country risk affects stock market returns and volatility (Suleman et al., 2017).

The international business literature has established the association between country risk and firm performance (Goldschmidt et al., 2011). Multinational enterprises have to dynamically consider local conditions worldwide while they adapt their optimal investment strategies. In particular, they have to critically assess any changes in the political environments that may cause political risks (Kesternich and Schnitzer, 2010). Managers tend to minimize uncertainty and rely on country risk assessments for formulating effective strategies. In the literature, there is

consensus that country risk is a function of political, social, and economic factors. However, it is very difficult to establish causation in this context because the risk dimensions are highly correlated and mutually interdependent (Cavusgil et al., 2020; Oetzel et al., 2001). For managers, it is important to understand the impact of future economic and political events on the business climate and, therefore, they need to rely on country risk assessments as a crucial aspect of strategic decision-making (Oetzel et al., 2001).

Amankwah-Amoah and Wang (2019) defined business risk as “threats posed to the focal business stemming from the political, economic, societal and technological environment of the firm”. It is well established in the literature that the management of business risk is critical for the success of any enterprise. Both qualitative and quantitative approaches have been explored in the literature to measure the probability and severity of adverse events on business viability while considering the political, social, and economic aspects of risk. For instance, Qazi and Khan (2021) investigated dependencies among country risk drivers using a BBN model. They identified international transactions policy, legal system, and fiscal policy as the most critical factors that can impact different categories of country risk, including financial, economic, and political risks.

2.3. Association between competitiveness and country risk

Although several studies have explored dependencies among individual country risk drivers (Qazi, 2022; Qazi and Khan, 2021) and global competitiveness pillars (Cinicioglu et al., 2017; Önsel Ekici et al., 2016) for prioritizing factors that influence country risk and competitiveness, respectively, the association between country risk and competitiveness remains underexplored. Given the challenge of directly assessing country risk drivers and forecasting country risk (Doumpos and Zopounidis, 2001; Li et al., 2012; Qazi and Khan, 2021), the GCI 4.0 framework emerges as a potential tool for policymakers and credit rating agencies to predict and evaluate country risk. Moreover, the GCI 4.0 assessments are grounded in subjective estimates, and juxtaposing these with country risk-related data might serve to enhance the accuracy and comprehensiveness of the GCI 4.0's country-level evaluations. By associating the GCI 4.0's nuanced insights into a country's competitiveness with its susceptibility to risk factors, decision-makers can better inform policy formulation, investment decisions, and strategic planning (Cinicioglu et al., 2017).

Exploring the association between GCI 4.0 and country risk is closely related to technological forecasting due to the profound influence of technological advancements on both competitiveness and risk factors within nations (Qazi and Al-Mhdawi, 2023; WEF, 2019b). The GCI 4.0 framework emphasizes key drivers of productivity that are deeply intertwined with the Fourth Industrial Revolution, driven by technological innovations (Samarah, 2023). As countries adopt and integrate emerging technologies like artificial intelligence, automation, and digital connectivity, their competitive strengths and weaknesses evolve (Asgary et al., 2020). This interaction between technological progress and competitiveness is a core focus of exploring the GCI-country risk association.

Furthermore, technological forecasting plays a pivotal role in assessing and managing country risk (Qazi and Al-Mhdawi, 2023). As countries embrace digital transformation, their economic, social, and political landscapes are reshaped. Innovations can mitigate some risk dimensions while introducing new ones, such as cyber threats or job displacement due to automation (Aly, 2022). Understanding how a nation's technological capabilities intersect with its exposure to various risk factors is critical for policymakers, businesses, and investors to anticipate and prepare for potential disruptions (Tamvada et al., 2022).

This study thus aims to bridge this knowledge gap by delving into the intricate relationship between GCI 4.0 and country risk. By shedding light on how a nation's competitive strengths and vulnerabilities intersect, this research holds the potential to offer novel insights that can

guide policymakers, investors, and stakeholders in making informed decisions that balance economic growth with risk mitigation and sustainable development.

3. Research methodology

The objective of this study could be achieved using several techniques, such as regression analysis, time-series analysis, machine learning techniques, structural equation modeling, ANNs, and others. A BBN modeling approach was utilized due to the efficacy of BBNs in modeling risk and uncertainty (Cox et al., 2018). In particular, BBNs offer several advantages over other probabilistic techniques, making them a preferred choice for this particular study. First, BBNs provide an intuitive and visual representation of the relationships between variables in the form of a graph, making it particularly useful for communication and decision-making (Hossain et al., 2020). Second, BBNs naturally capture cause-and-effect and statistical relationships between variables, enabling the model to capture how changes in one variable can affect others (Qazi et al., 2022). Third, BBNs allow for efficient probabilistic inference, meaning they can calculate the probabilities of different outcomes based on observed evidence (Qazi et al., 2021). This makes them suitable for predicting future events or assessing the likelihood of different scenarios. Fourth, BBNs can assist in feature selection by highlighting the most relevant variables in the network (Al Nuairi et al., 2023). This can be valuable for understanding the key drivers of certain outcomes or for optimizing models in terms of complexity and performance. Moreover, BBNs are used in decision support systems to aid in making informed choices under uncertainty (Hosseini and Ivanov, 2020). They can model the potential outcomes of different decisions and help select the most favorable course of action.

A two-stage methodology was adopted to achieve the research objective: modeling and validation of two data-driven BBN models using country-level data related to country risk and GCI 4.0; and prioritization of factors in a network setting. Delen et al. (2020) proposed a methodology to identify critical variables that can influence student attrition. The proposed methodology can effectively capture dependencies among variables and therefore, a similar approach was adopted in this study.

This study utilized the data-sets by AM Best (AMBEST, 2019) and the World Bank (WEF, 2019a) regarding country risk and GCI ratings, respectively. The GCI 4.0 is based on different indicators derived from a survey and WEF indexes, such as the Global Gender Gap Index and the Travel and Tourism Competitiveness Index. Several publications, such as the Inclusive Growth and Development Report and Global Risks Report, are also used as supplementary resources. The 2019 edition of the survey involved 16,936 business executives in 139 economies between January and April 2019. The survey was administered in 41 languages. A total of 12,987 responses were retained and more than half of the retained surveys were completed online (WEF, 2019a). The survey comprised 78 questions to assess the performance of various domains of individual countries using a scale of 1 (considered among the worst in the world) to 7 (considered among the best in the world). The questions were organized into 10 topical areas: infrastructure; technology; financial environment; foreign trade and investment; domestic competition; business operations and innovation; security; governance; education and human capital; and risks (WEF, 2019a).

In total, 125 and 141 countries were represented in the data-set for country risk and GCI assessments, respectively. However, the consolidated data-set comprised the assessments for 107 countries due to the different countries reflected in the two data-sets. All 12 GCI 4.0 pillars and 12 country risk drivers were retained in the consolidated data-set. The GCI 4.0-related data-set utilized a continuous scale of 0–100 for assessing each of the 12 pillars (see Table I for descriptive statistics), whereas the country risk-related data-set used a discrete scale of 1–5 (see Table II for descriptive statistics).

BBNs, also known as Bayesian networks or probabilistic graphical models, are a powerful tool used in probability theory and statistics to

Table I
Descriptive statistics of the GCI 4.0 pillars and country risk.

Variable	Mean	Var	St. dev	Min	Max	Count
Business dynamism	62.5	103.1	10.2	29	84	107
Country risk	3.4	1.9	1.4	1	5	107
Financial system	66.4	178.4	13.4	39	91	107
Health	80.9	218.5	14.8	33	100	107
ICT adoption	59.8	286.9	16.9	20	93	107
Infrastructure	70.3	190.4	13.8	35	95	107
Innovation capability	46.9	295.5	17.2	22	87	107
Institutions	57.3	137.4	11.7	26	81	107
Labor market	61.1	73.5	8.6	43	81	107
Macroeconomic stability	83.1	296.5	17.2	0	100	107
Market size	59.7	247.7	15.7	19	100	107
Product market	56.9	70.7	8.4	36	82	107
Skills	64.7	156.1	12.5	30	87	107

Table II
Descriptive statistics of country risk drivers and GCI 4.0.

Variable	Mean	Var	St. dev	Min	Max	Count
Business environment	3.0	0.8	0.9	1	5	107
Economic risk	3.1	1.1	1.0	1	5	107
Financial system risk	3.3	2.0	1.4	1	5	107
Fiscal policy	3.3	1.1	1.0	1	5	107
GCI 4.0	64.1	121.8	11.0	38	85	107
Government stability	3.2	0.8	0.9	2	5	107
International transactions policy	2.9	1.7	1.3	1	5	107
Labor flexibility	3.4	0.7	0.8	2	5	107
Legal system	3.4	1.1	1.0	1	5	107
Monetary policy	2.3	0.5	0.7	1	5	107
Political risk	3.0	1.5	1.2	1	5	107
Regional stability	3.2	2.0	1.4	1	5	107
Social stability	3.0	0.7	0.8	2	5	107

model and analyze uncertain and complex systems (Yang et al., 2022). BBNs provide a structured and visual way to represent relationships between variables, their dependencies, and the probabilities of different outcomes (Gaudenzi and Qazi, 2021). At the core of a BBN is a directed acyclic graph (DAG), where nodes represent variables and edges represent probabilistic dependencies between the variables (Barry, 2021). The direction of the edges indicates the causal relationships or statistical dependence between variables (Marcot and Hanea, 2021). An edge from node A to node B suggests that A influences B. In other words, the occurrence of A might impact the occurrence of B. This is used to represent conditional dependencies between variables. Each node has an associated conditional probability table (CPT) that specifies the probabilities of its possible values given the values of its parent nodes. The CPT quantifies the strength of influence between nodes and is a crucial component of a BBN (Hossain et al., 2020).

BBNs enable inference through a process called probabilistic inference or belief propagation (Adedipe et al., 2020). Given certain evidence or observations (values assigned to specific nodes), the network can calculate the probabilities of other variables' states. This is useful for making predictions or assessing the likelihood of different outcomes in uncertain scenarios (Lin et al., 2020). BBNs can be learned from data using statistical techniques (Cox et al., 2018). Learning involves estimating the parameters of the CPTs from observed data, effectively adapting the network to real-world scenarios. BBNs find applications in various fields such as finance (Qazi and Simsekler, 2022), healthcare (Al Nuairi et al., 2023), engineering (Hossain et al., 2020), and others (Hosseini and Ivanov, 2020). They are used for decision-making, risk assessment, prediction, diagnosis, and scenario analysis (Adedipe et al., 2020).

Both historical data and expert judgment can be used to develop BBNs (Hanea et al., 2018). This paper explores a data-driven approach to assess dependencies between country risk and GCI 4.0. The data-driven

discrete BBN modeling approach involves establishing discrete states for individual variables, developing a qualitative network structure, and learning network parameters that represent the strength of dependency between interconnected variables. Several software packages, such as GeNIe and AgenaRisk, can be used to develop a BBN model (Qazi and Khan, 2021). Kjaerulff and Madsen (2008), Jensen and Nielsen (2007), and Cox et al. (2018) provide a detailed description of the mechanics of BBNs and the use of probability theory in developing and analyzing BBN models.

A three-state discretization scheme was adopted for all variables as per prior studies (de Oliveira et al., 2012; Simsekler and Qazi, 2022). The data were processed in GeNIe 2.0. Interested readers may consult BayesFusion (2017) for a comprehensive overview of the processing of data and discretization schemes available in the GeNIe software for developing a BBN model. The country risk drivers were discretized into three states such that s1 represented a low level of country risk (a score of 1 or 2), s2 a medium level (a score of 3), and s3 a high level of risk (a score of 4 or 5). On the other hand, the GCI ratings were discretized into s1 (a score of up to 50), s2 (a score between 50 and 70), and s3 (a score above 70). A uniform-width discretization scheme could not be adopted due to the unsymmetrical distribution related to the GCI ratings (see Table I). Based on the classification of states for the GCI pillars and country risk drivers, s1 (s3) is the most preferred state for individual country risk drivers (GCI pillars).

Using the discretized data for the variables (see Tables I and II), two BBN models were developed in GeNIe 2.0 (see Figs. 1a and 1b). The Tree-augmented Naive (TAN) BBN learning algorithm was used to develop the models because this algorithm is effectively utilized in other application areas for predicting different variables of interest (Delen et al., 2020; Önsel Ekici et al., 2019). The TAN models allow for interdependencies among predictor variables in addition to the direct association between individual predictor variables and the target variable (Cox et al., 2018). The BBN models were validated using the k-fold cross-validation technique (Marcot and Hanea, 2021). K-fold cross-validation is a technique used to evaluate the performance and generalization ability of machine learning models, including BBNs (Zhou et al., 2023). It involves splitting the available data-set into K subsets or "folds" and performing a series of iterations, where the model is trained on K-1 folds and tested on the remaining fold. K-fold cross-validation provides a more robust estimate of a model's performance compared to a single train-test split (Chen and Zhang, 2021). The models were able to achieve an accuracy level of 81.3 % and 82.2 % for the prediction of GCI 4.0 and country risk, respectively.

The results of the two models and the prioritization of factors relative to GCI 4.0 and country risk are discussed in the next section.

4. Results and interpretations

4.1. BBN model exploring the country risk-GCI pillars nexus

The structural model of TAN is illustrated in Fig. 1a, depicting the individual factors as nodes interconnected by directed arcs to represent their interdependencies. In the TAN algorithm, a specific target variable is determined, such as country risk in our context. The fundamental assumption is that all the factors maintain a direct connection with the target variable, which in this case is country risk. This rationale clarifies the presence of an arc originating from the country risk node and extending toward each of the other nodes. Nevertheless, it is important to comprehend this relationship in a way that underscores how all the factors (GCI 4.0 pillars) exert influence on the target variable (country risk), rather than the reverse (Delen et al., 2020). Notably, the TAN algorithm permits interactions among the nodes while confining the number of parent nodes for each node to two, which includes the target variable itself (Al Nuairi et al., 2023).

Besides depicting direct cause-effect relationships between individual GCI pillars and country risk, Fig. 1a also indicates direct and indirect

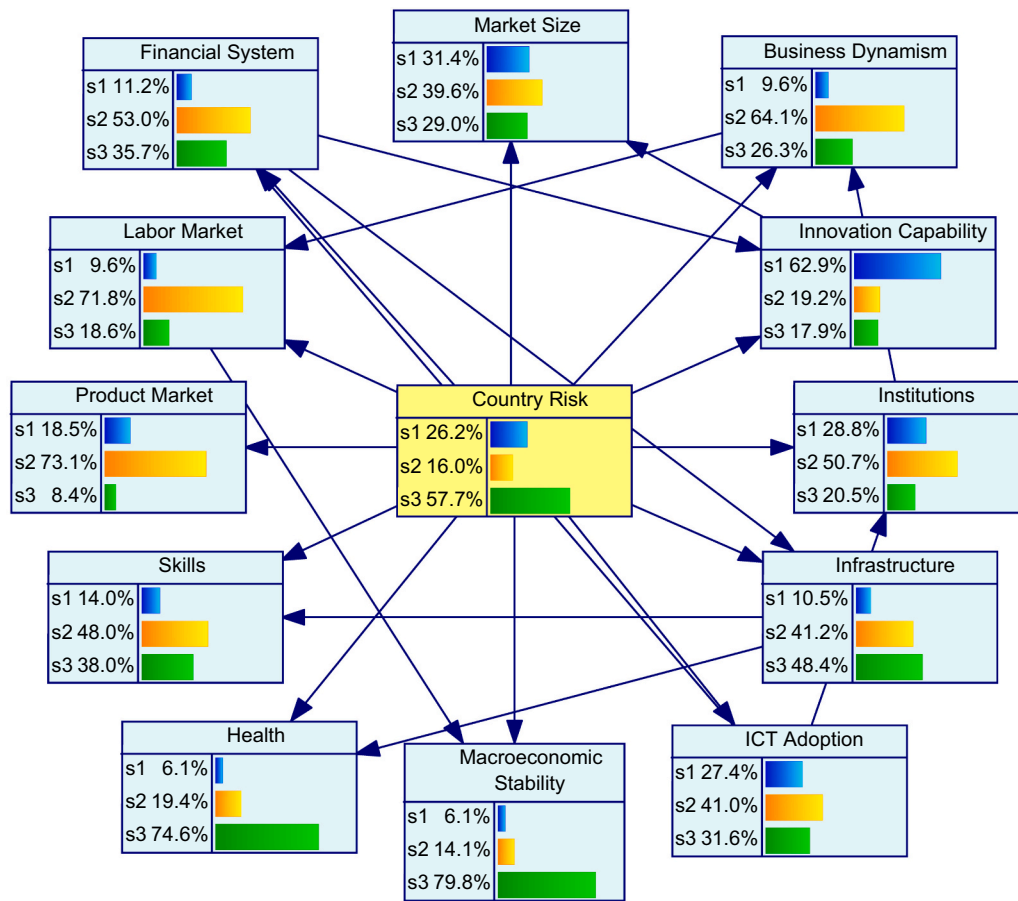


Fig. 1a. A BBN model representing relations among the GCI 4.0 pillars and country risk.

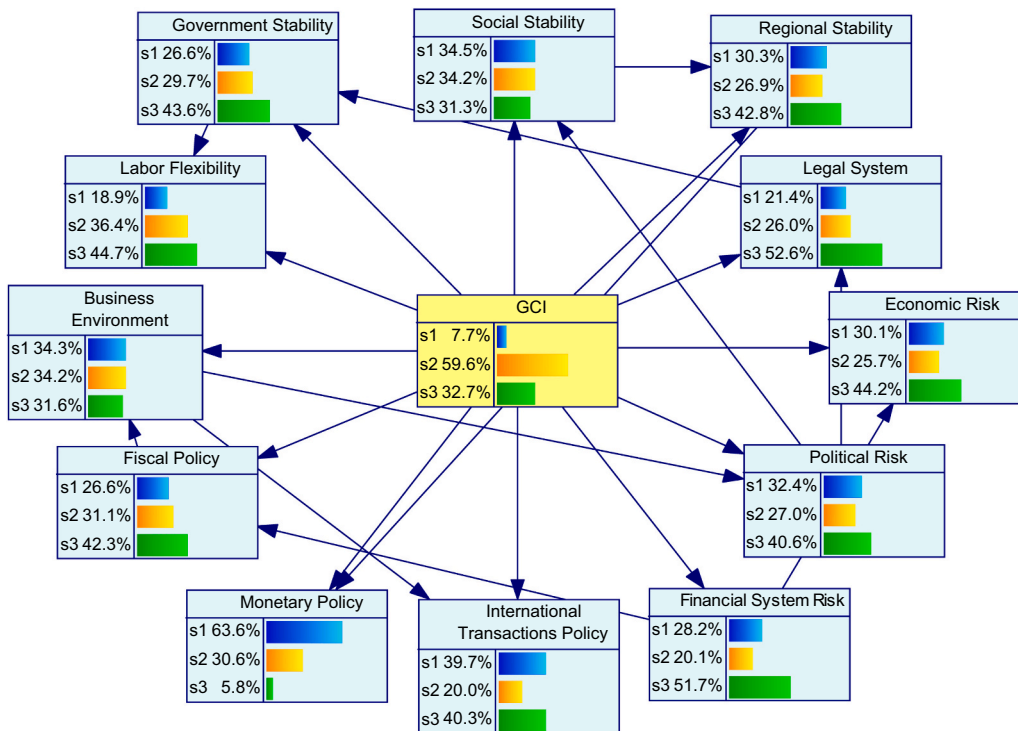


Fig. 1b. A BBN model representing relations among country risk drivers and GCI 4.0.

associations among the GCI pillars. For instance, the financial system directly influences innovation capability and infrastructure. Infrastructure, in turn, influences skills and health, whereas innovation capability affects market size. Therefore, the financial system is directly or indirectly related to multiple GCI pillars. ICT adoption directly influences the financial system and institutions. Similarly, business dynamism affects the labor market, whereas institutions influence business dynamism and the product market.

The BBN model (see Fig. 1a) also presents the probability distribution for each variable in the form of a bar chart. For instance, the probability distribution of innovation capability is negatively skewed with a high proportion of countries (i.e., 62.9 %) associated with the low (performance) state. On the other hand, macroeconomic stability- and health-related distributions are positively skewed. In terms of country risk, more than half of the countries (i.e., 57.7 %) are associated with the high-risk state.

4.2. BBN model exploring the GCI 4.0-country risk drivers nexus

Fig. 1b represents dependencies between the GCI 4.0 and country risk drivers. Political risk directly influences the legal system and social stability. Social stability, in turn, influences regional stability. Similarly, financial system risk directly affects economic risk and fiscal policy, whereas fiscal policy influences the business environment. The legal system influences government stability, which further affects labor flexibility. The probability distribution of monetary policy is negatively skewed with a high proportion of countries (i.e., 63.6 %) associated with the s1 state. On the other hand, financial system risk- and legal system-related distributions are positively skewed. In terms of the GCI, more than half of the countries (i.e., 59.6 %) are associated with the medium (performance) state (i.e., s2).

4.3. Diagnostic value of GCI 4.0 pillars and country risk drivers

The GCI 4.0 pillars and country risk drivers were ranked based on their diagnostic value in predicting a certain state of country risk and GCI 4.0, respectively (see Figs. 2 and 3). The ranking is based on an information-theoretic measure known as cross-entropy (Delen et al., 2020) and expresses, for each GCI pillar (country risk driver) individually, the expected reduction in entropy of the probability distribution of country risk (GCI) after observing the value of the GCI pillar (country risk driver). Cross-entropy is a utility-free measure of the value of information (BayesFusion, 2017) and it gives a good idea about the value of the GCI pillars (country risk drivers) for diagnosing the true state of country risk (GCI).

Among the diagnostic assessments for individual states of country risk, the minimum values were observed for the s2 state (see Fig. 2 (b)). Innovation capability and skills are the most critical pillars that can help predict country risk. On the other hand, market size, product market, health, and macroeconomic stability are the least critical pillars. Certain pillars differ in their importance relative to the two extreme states of country risk. For instance, institutions seem to be important for the s1 state of country risk, whereas these are relatively less important for the s3 state of country risk.

Among the diagnostic assessments for individual states of GCI 4.0, the minimum values were observed for the s1 state (see Fig. 3 (a)). Business environment and financial system risk are the most critical country risk drivers that can help predict GCI 4.0. On the other hand, labor flexibility, regional stability, and monetary policy are the least critical pillars. All country risk drivers possess a similar ranking relative to predicting the s2 and s3 states of GCI 4.0 (see Fig. 3 (b) and (c)).

4.4. Impact assessment of GCI 4.0 pillars and country risk drivers

Sensitivity analysis was performed to understand the impact of different GCI pillars on country risk. First, the worst possible state of

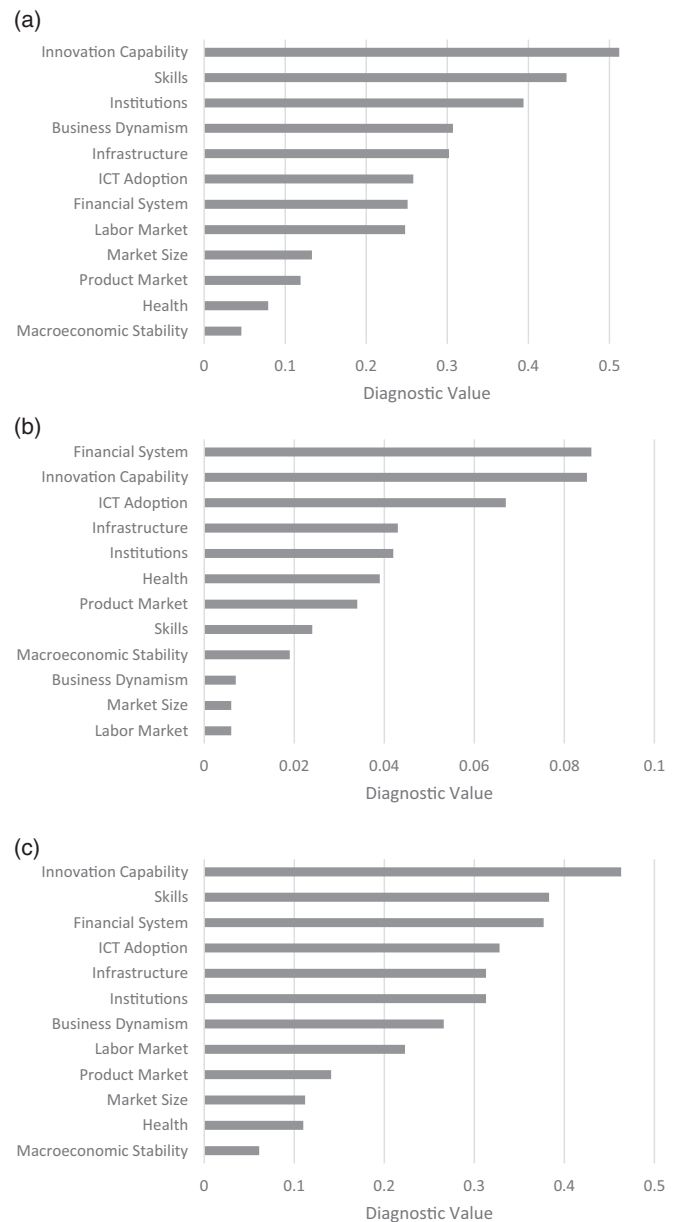


Fig. 2. Diagnostic value of the GCI 4.0 pillars for predicting the (a) s1 state; (b) s2 state; (c) s3 state of country risk.

individual GCI pillars was selected for the model (see Fig. 1a) and the corresponding impact on country risk was ascertained (see Fig. 4 (a)). ICT adoption is the most critical GCI pillar that can impact country risk yielding a probability of 94.6 % for the high-risk state. Similarly, product market, institutions, and skills are other important pillars. On the other hand, health and macroeconomic stability are the least critical pillars. Second, the most desirable state of individual GCI 4.0 pillars was selected for the model (see Fig. 1a) and the corresponding impact on country risk was determined (see Fig. 4 (b)). Innovation capability is the most critical GCI 4.0 pillar that can reduce country risk with a probability of 86.1 % for the low-risk state. Similarly, institutions and the labor market are other important pillars. On the other hand, health and macroeconomic stability remain the least critical pillars.

Sensitivity analysis was also performed to understand the impact of different country risk drivers on GCI 4.0. First, the most desirable state of individual country risk drivers was selected for the model (see Fig. 1b) and the corresponding impact on GCI 4.0 was ascertained (see Fig. 5 (a)). The legal system is the most critical country risk driver that can

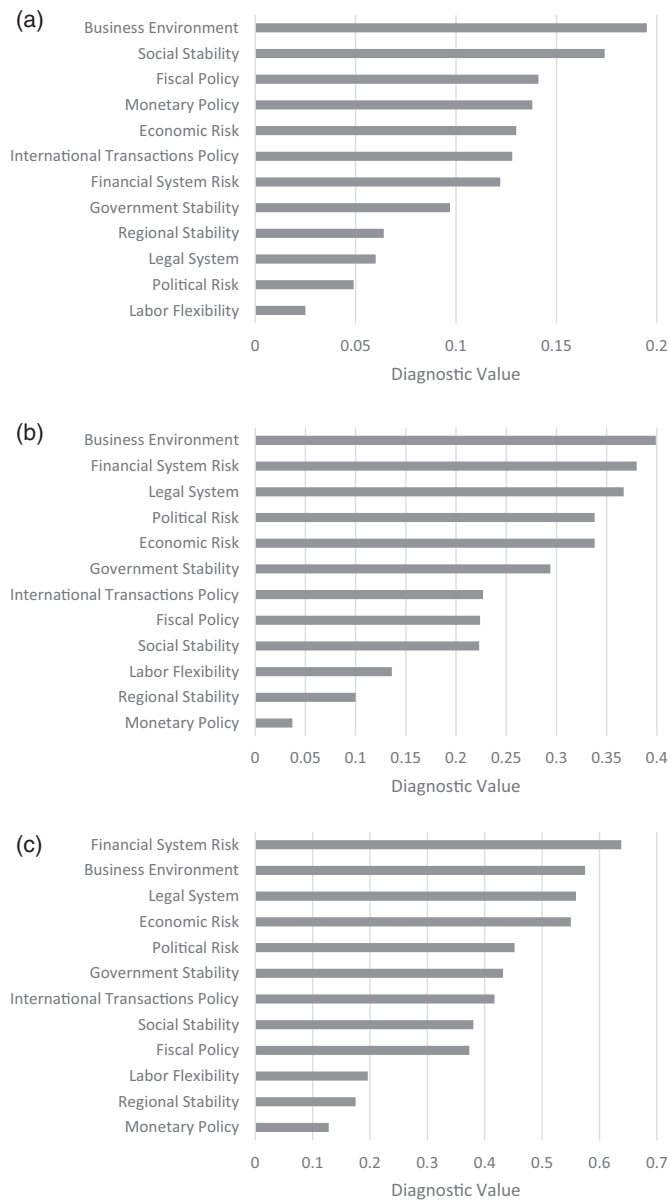


Fig. 3. Diagnostic value of country risk drivers for predicting the (a) s1 state; (b) s2 state; (c) s3 state of GCI 4.0.

impact GCI 4.0 yielding a probability of 92.7 % for the high (performance) state. Similarly, financial system risk, economic risk, government stability, and business environment are other important drivers. On the other hand, monetary policy is the least critical driver. Second, the least desirable state of individual country risk drivers was selected for the model (see Fig. 1b) and the corresponding impact on GCI 4.0 was determined (see Fig. 5 (b)). Monetary policy is the most critical country risk driver that can reduce GCI 4.0 with a probability of 24 % for the low (performance) state. Similarly, business environment and social stability are other important drivers. On the other hand, labor flexibility is the least critical driver with a probability of 11 % for the low (performance) state.

4.5. Ranking of GCI 4.0 pillars and country risk drivers

The two ranking schemes of the GCI 4.0 pillars were analyzed based on their positive and negative impact on country risk corresponding to the two extreme (performance) states (see Table IIIa). Institutions were relatively important in both ranking schemes. Market size, health, and

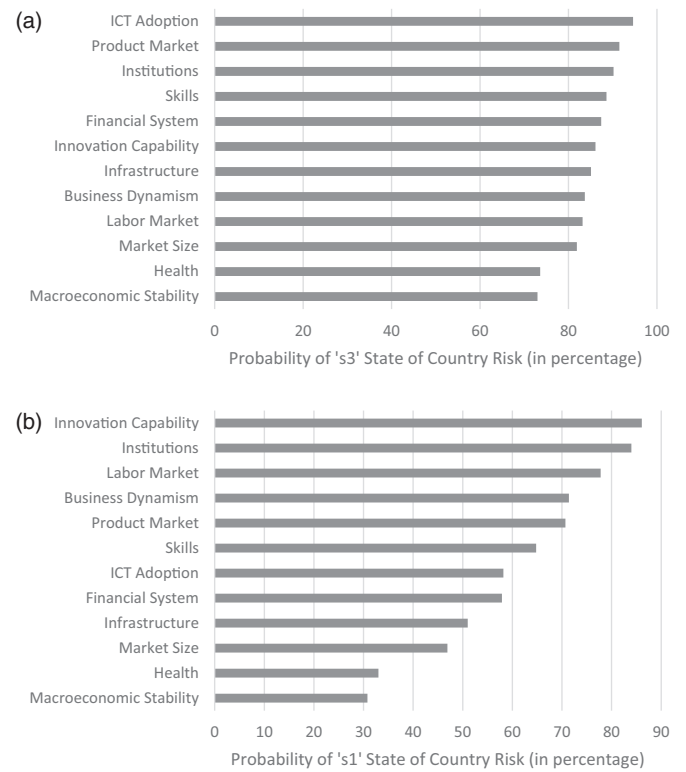


Fig. 4. Impact assessment of the (a) worst state; (b) best state of individual GCI 4.0 pillars on country risk.

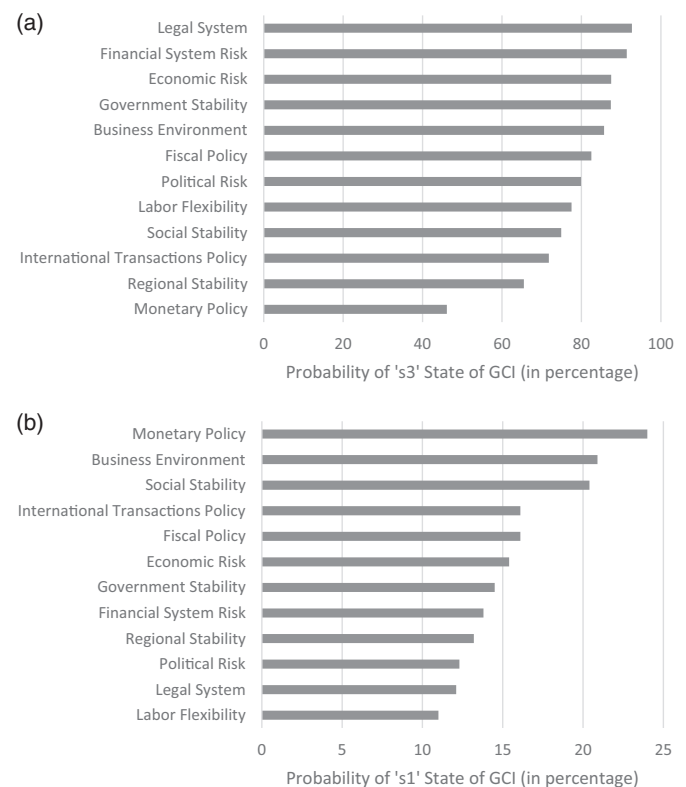


Fig. 5. Impact assessment of the (a) most desirable state; (b) least desirable state of individual country risk drivers on GCI 4.0.

Table IIIa

Ranking of the GCI 4.0 pillars based on their impact on country risk for the two extreme (performance) states.

GCI 4.0 pillar	Ranking relative to 's1' state	Ranking relative to 's3' state
Business dynamism	8	4
Financial system	5	8
Health	11	11
ICT adoption	1	7
Infrastructure	7	9
Innovation capability	6	1
Institutions	3	2
Labor market	9	3
Macroeconomic stability	12	12
Market size	10	10
Product market	2	5
Skills	4	6
Correlation	0.56	
p-Value	0.059	

macroeconomic stability were the least critical pillars in both schemes. However, there were some pillars whose ranking changed significantly across the two schemes, such as ICT adoption, innovation capability, and labor market. There was a moderate correlation between the two ranking schemes with a *p*-value of 0.059.

Similarly, the two ranking schemes of country risk drivers were analyzed based on their positive and negative impact on GCI 4.0 corresponding to the two extreme (risk) states (see Table IIIb). There was a negative weak correlation between the two ranking schemes, but it was statistically non-significant. There were significant differences between the two ranking schemes. For instance, the legal system and monetary policy were the most and least important drivers in the first scheme, respectively; however, it was the opposite in the second scheme. Similarly, there was a wide variation in the ranking of financial system risk, social stability, and international transactions policy across the two schemes. Labor flexibility, political risk, and regional stability were relatively non-critical drivers in both schemes.

5. Discussion and implications

Önsel Ekici et al. (2019) identified innovation as a critical GCI pillar that can help improve economic development, especially country-level logistics performance. Similarly, Dima et al. (2018) identified innovation and education as critical determinants of the competitiveness and economic development of the European Union. However, this study establishes the importance of innovation capability in predicting country risk. Human capital is considered an important factor in the fourth

Table IIIb

. Ranking of country risk drivers based on their impact on GCI 4.0 for the two extreme (risk) states.

Country risk driver	Ranking relative to 's1' state	Ranking relative to 's3' state
Business environment	5	2
Economic risk	3	6
Financial system risk	2	8
Fiscal policy	6	4
Government stability	4	7
International transactions policy	10	4
Labor flexibility	8	12
Legal system	1	11
Monetary policy	12	1
Political risk	7	10
Regional stability	11	9
Social stability	9	3
Correlation	−0.34	
p-Value	0.285	

industrial revolution era (WEF, 2019a); therefore, skills are assessed as an important pillar that can help predict country risk.

Business environment and financial system risk are critical drivers of country risk (Qazi and Khan, 2021). Several studies have established the importance of these drivers (Cavusgil et al., 2020; Deligonul, 2020; Gregory, 2020). However, it remains underexplored how these drivers can help predict GCI 4.0. This study identifies business environment and financial system risk as the most critical drivers to forecast GCI 4.0. These findings can help policymakers and researchers develop better forecasts regarding country risk and GCI 4.0 assessments and validate their results.

ICT adoption is the most critical GCI pillar that can negatively impact country risk. Therefore, policymakers need to develop contingency plans. The importance of ICT adoption is established in other studies as well (Wu et al., 2012). However, this study emphasizes its vulnerability potential, implying that this pillar may not be the most critical pillar for reducing country risk. Moreover, contingency plans may also be developed regarding other critical pillars, including product market, institutions, and skills. While reducing country risk, policymakers may need to focus on innovation capability. Similarly, limited resources may be allocated to other critical pillars, including institutions and the labor market.

Policymakers, who aim to improve GCI 4.0, need to prioritize legal system, financial system risk, economic risk, government stability, and business environment. On the other hand, contingency plans may have to be developed regarding monetary policy, business environment, and social stability due to the significant negative impact of these country risk drivers on GCI 4.0. The relative importance of individual GCI pillars and country risk drivers established in this study can be utilized to allocate limited resources to critical pillars and drivers.

The proposed modeling approach can be utilized to examine the variation in competitiveness and risk across different states within a country. This can help policymakers identify critical regions for resource allocation to improve productivity and resilience against various financial, economic and political vulnerabilities. Moreover, the graphical interface can help stakeholders prioritize potential strategies and visualize the impact of their actions on the interconnected web of competitiveness pillars and country risk drivers.

Exploring the association between GCI 4.0 and country risk holds significant theoretical implications that contribute to our understanding of the intricate relationship between a nation's competitive capabilities and its vulnerability to various risk factors. In particular, the models presented provide an opportunity to explore the two-way interaction between competitiveness and country risk, thereby making it possible to improve the prediction of these constructs using a consolidated list of relevant factors. This study also sheds light on how a country's competitiveness can act as both a shield against adverse influences and a catalyst for growth, potentially mitigating risk factors. This exploration not only enhances our understanding of the complex interplay between economic dynamism and risk management but also offers insights into the strategies that countries can adopt to navigate a global landscape characterized by uncertainty and rapid technological change.

Exploring the association between the GCI 4.0 and country risk can inform strategic decision-making for businesses and policymakers alike. A deeper understanding of how a nation's competitiveness and its exposure to risk factors are interconnected can guide multinational enterprises in formulating more informed international expansion strategies. Managers can leverage insights from this study to tailor their investment decisions, supply chain management, and market entry approaches based on a country's competitive strengths and risk profile. Moreover, policymakers can use the findings of this study to design targeted policies that foster innovation, build resilient institutions, and enhance human capital development—critical factors that not only contribute to competitiveness but also help buffer a country against potential risk events. By acknowledging the interrelation between competitiveness and risk (see Figs. 1a and 1b), managers can adopt a

more holistic approach to risk assessment and mitigation, leading to more effective risk management strategies and sustainable business growth in an increasingly dynamic global economy.

This study has made two unique contributions to the literature on country risk analysis and global competitiveness. First, this is the first study to establish the association between country risk and global competitiveness in a probabilistic network setting. Second, this study operationalizes a data-driven BBN approach in the form of an innovative application, which can help researchers and practitioners establish the relative importance of country risk drivers and GCI pillars that can influence competitiveness and country risk, respectively. This study provides an opportunity to explore global competitiveness through the lens of risk science (Aven, 2004, 2016a,b, 2017). Although risk management has been explored in different areas of economic development, including logistics and supply chain management (Choi, 2021; Hosseini et al., 2019; Wan et al., 2019), its application in the context of competitiveness is fragmented. In particular, the association between country risk and global competitiveness remains underexplored.

6. Conclusions

This study explored the prediction of global competitiveness and country risk using the country risk drivers and GCI 4.0 pillars, respectively. Among the GCI pillars, innovation capability and macroeconomic stability are associated with the highest and lowest diagnostic values relative to the prediction of country risk, respectively. On the other hand, among the country risk drivers, financial system risk and monetary policy are associated with the highest and lowest diagnostic value relative to the prediction of GCI, respectively. Scenario analysis revealed that low performance in monetary policy can significantly affect GCI 4.0, whereas financial system risk and legal system can considerably improve GCI 4.0. On the other hand, low performance in ICT adoption can significantly increase country risk, whereas innovation capability and institutions can considerably reduce country risk.

This study has certain limitations. The sensitivity of the BBN models developed in this paper has not been performed against different discretization schemes and algorithms. Indicators and factors related to the GCI pillars and country risk drivers have not been modeled. Potential strategies that can address critical risk drivers and improve the performance of individual GCI pillars are not considered. Future research can benefit from exploring dependencies among individual country risk drivers and GCI pillars. BBN models can be developed using data from multiple years. The Augmented Naive Bayes (ANB) algorithm can be used to explore causation among multiple GCI pillars and country risk drivers. Finally, a comprehensive causal network can be developed to include additional risk factors and potential risk mitigation strategies.

Data availability

Data will be made available on request.

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